



Combining probability forecasts

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Summary. Linear pooling is by far the most popular method for combining probability forecasts. However, any non-trivial weighted average of two or more distinct, calibrated probability forecasts is necessarily uncalibrated and lacks sharpness. In view of this, linear pooling requires recalibration, even in the ideal case in which the individual forecasts are calibrated. Towards this end, we propose a beta-transformed linear opinion pool for the aggregation of probability forecasts from distinct, calibrated or uncalibrated sources. The method fits an optimal non-linearly recalibrated forecast combination, by compositing a beta transform and the traditional linear opinion pool. The technique is illustrated in a simulation example and in a case-study on statistical and National Weather Service probability of precipitation forecasts.

Keywords: Calibration; Coherent combination formula; Forecast aggregation; Linear opinion pool; Model averaging; Probability forecasting; Reliability; Resolution; Sharpness

1. Introduction

Probabilistic forecasts take account of the uncertainty in a prediction, by taking the form of a predictive probability distribution for a future quantity or event. The simplest case is that of a future binary or dichotomous event, such as a recession *versus* no recession, or rain *versus* no rain. In the binary case, a predictive probability distribution is simply an *ex ante* probability for the event to happen. Although the roots of probability forecasting can be traced back to the 18th century, the transition to probability of precipitation forecasts by the US National Weather Service (NWS) in 1965 was perhaps the most influential and important event in their development (Murphy, 1998; Winkler and Jose, 2008). In economics, the Survey of Professional Forecasters has included probability variables since 1968 (Croushore, 1993). Of course, there are many other important applications of probability forecasts, including but not limited to medical diagnosis (Spiegelhalter, 1986), educational testing and political and socio-economic foresight (Tetlock, 2005). Arguably, a far-reaching transdisciplinary transition to distributional forecasting is well under way (Gneiting, 2008).

In many instances, multiple probability forecasts for the same event are available. In surveys, economic experts might provide diverse probability assessments of a future recession. Distinct numerical and/or statistical models might provide a collection of probability of precipitation forecasts, and a group of physicians might assign individual survival probabilities. In this type of situation, there is strong empirical evidence that combined probability forecasts that draw on all

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the experts' or models' strengths result in improved predictive performance. This is very much in the spirit of model averaging, which has primarily been developed for statistical inference (Hoeting *et al.*, 1999).

Various ways of combining probability forecasts into a single aggregated forecast have been proposed. Genest and Zidek (1986), Wallsten *et al.* (1997), Clemen and Winkler (2007) and Primo *et al.* (2009) provide excellent reviews. In practice, most aggregation techniques rely on a weighted linear combination of the individual probability forecasts, which is often referred to as a linear opinion pool (Stone, 1961; Genest and McConway, 1990; DeGroot and Mortera, 1991). Substantial empirical evidence attests to the benefits of linear opinion pools, with successful applications ranging from meteorology (Sanders, 1963; Vislocky and Fritsch, 1995; Baars and Mass, 2005) to economics (Graham, 1996), psychology (Ariely *et al.*, 2000) and medical diagnosis (Winkler and Poses, 1993), among other fields.

The goal in probability forecasting is to maximize the sharpness of the forecast distributions subject to calibration (Murphy and Winkler, 1987; Gneiting *et al.*, 2007; Pal, 2009). Calibration or reliability measures how close conditional event frequencies are to the forecast probabilities. Sharpness describes how far away the forecasts are from the naive, climatological baseline forecast, i.e. the marginal event frequency (Gneiting *et al.*, 2008; Winkler and Jose, 2008). The more extreme the forecast probabilities are, i.e. the closer to the most confident values of 0 or 1, the sharper the forecast. Strictly proper scoring rules such as the Brier or quadratic score (Brier, 1950; Selten, 1998) and the logarithmic score (Good, 1952) provide summary measures of predictive performance that address calibration and sharpness simultaneously (Gneiting and Raftery, 2007).

It is therefore critical that probability assessments are aggregated in ways that promote calibrated and sharp combined forecasts. In Section 2 we demonstrate a striking result, in that any weighted linear combination of distinct, individually calibrated probability forecasts is necessarily uncalibrated and lacks sharpness. In this light, linear opinion pools are suboptimal, so in Section 3 we propose a non-linear generalization, the beta-transformed linear pool (BLP). The BLP method fits an optimally recalibrated forecast combination, by compositing a beta transform and the traditional linear opinion pool. Section 4 illustrates the BLP method in a case-study on statistical and NWS probability of precipitation forecasts at 29 major cities in continental USA. The BLP combined forecast is calibrated and sharp and outperforms the individual and linearly combined forecasts. The paper closes with a discussion in Section 5.

2. Some shortcomings of linearly combined probability forecasts

The overarching message in this section is that linear opinion pools are generally uncalibrated, even in the ideal case in which each individual source is calibrated. We give a rigorous probabilistic version of this result in theorem 1, which is then illustrated in a simulation study.

2.1. Theoretical results

We work within a probabilistic framework which considers the joint distribution of the random vector

$$(Y, p_1, \dots, p_k),$$

where $Y \in \{0, 1\}$ is a binary or dichotomous event, and $0 \leq p_1, \dots, p_k \leq 1$ are probability forecasts that take values in the closed unit interval. This is akin to the setting in DeGroot and Fienberg (1982, 1983) and Murphy and Winkler (1987) but considers an arbitrary number k of individual probability forecasts, each of which is a random variable, with full generality in the joint depen-

dence structure. In this framework a probability forecast is any random variable p that is measurable with respect to the σ -algebra that is generated by $p_1, \dots, p_k$, with the linear opinion pool,

$$p = w_1 p_1 + \dots + w_k p_k, \quad w_1, \dots, w_k \geq 0, \quad w_1 + \dots + w_k = 1, \quad (1)$$

being one such example. The probability forecast p is calibrated for Y if

$$\mathbb{P}(Y = 1|p) = \mathbb{E}(Y|p) = p \quad \text{almost surely.}$$

This definition is in accordance with the economic, meteorological, psychological and statistical forecasting literature and can be traced to Murphy and Winkler (1987) and Schervish (1989). It differs from the game theoretic approach to calibration that has been developed in a far-reaching related strand of literature (Dawid, 1982; Foster and Vohra, 1998; Lehrer, 2001; Sandroni *et al.*, 2003; Vovk and Shafer, 2005). From the basic properties of conditional expectations, it is immediate that if p is a calibrated probability forecast then

$$\mathbb{E}(p) = \mathbb{E}\{\mathbb{E}(Y|p)\} = \mathbb{E}(Y).$$

This property can be thought of as a weak form of calibration, and we refer to it as marginal consistency. It resembles the notion of marginal calibration for probabilistic forecasts of continuous variables (Gneiting *et al.*, 2007).

A scoring rule assigns a numerical score $S(x, y)$ to the probability forecast $x \in [0, 1]$ and the binary event y , where $y = 1$ if the event occurs and $y = 0$ otherwise. We consider scoring rules to be negatively oriented penalties, i.e. the smaller the better. A scoring rule is strictly proper if it encourages honest assessments, i.e. if

$$x S(x, 1) + (1 - x) S(x, 0) < x S(x', 1) + (1 - x) S(x', 0) \quad \text{for all } 0 \leq x \neq x' \leq 1.$$

See Dawid (1986), Winkler (1996) and Gneiting and Raftery (2007) for reviews and discussion.

We now can state our key result. The proof is deferred to Appendix A.

Theorem 1. Suppose that $p_1, \dots, p_k$ are calibrated for the binary event Y and such that $p_i \neq p_j$ with strictly positive probability for at least one pair $i \neq j$. Consider the linear opinion pool

$$p = w_1 p_1 + \dots + w_k p_k,$$

where $w_1, \dots, w_k > 0$ and $w_1 + \dots + w_k = 1$. Let

$$q = \mathbb{P}(Y = 1|p) = \mathbb{E}(Y|p)$$

denote the recalibrated version of p , i.e. the conditional probability of Y given p . Then the following results hold.

- (a) The linear opinion pool p lacks calibration, in that $q \neq p$ with strictly positive probability.
- (b) The linear opinion pool p lacks sharpness, in that

$$\mathbb{E}(p - p_0)^2 < \mathbb{E}(q - p_0)^2$$

where $p_0 = \mathbb{E}(p) = \mathbb{E}(q) = \mathbb{E}(Y)$. In words, both p and q are marginally consistent, but on average p is closer to its expectation, the naive climatological forecast p_0 , than its recalibrated version, q .

- (c) The recalibrated forecast q is calibrated, i.e. $\mathbb{P}(Y = 1|q) = q$ almost surely, and it outperforms p , in that

$$\mathbb{E}\{S(q, Y)\} < \mathbb{E}\{S(p, Y)\}$$

for every strictly proper scoring rule S .

The statement about the lack of calibration of the linear opinion pool in part (a) is our key result. It is counter to a natural belief that linear pools of calibrated probability forecasts are calibrated, as recently expressed by Iversen *et al.* (2008), page 899. A similar result which applies to the case of multiple density forecasts for a continuous quantity was proved by Hora (2004). This uses a very different mode of calibration, and there is no apparent way of deducing our result from Hora's, or vice versa.

The result in part (a) is an immediate consequence of the stronger statement in part (b), since the latter implies that p cannot equal q with probability 1. Part (b) hints at the nature of the departure from the recalibrated forecast q and is expressed in terms of the expected squared deviation from the climatological baseline probability p_0 . For a sharp forecast, the forecast probabilities are close to 0 or 1, so the larger this deviation the sharper the forecast (Murphy and Winkler, 1992; Gneiting *et al.*, 2008; Winkler and Jose, 2008). As a result, the linear opinion pool is underconfident. Part (c) demonstrates the superiority of the recalibrated forecast in terms of strictly proper scoring rules (Gneiting and Raftery, 2007) and is akin to theorem 6.3 of Schervish (1989). Proper scoring rules address calibration and sharpness simultaneously, so in view of parts (a) and (b) this is an unsurprising result.

We proceed to discuss related results in the literature. Dawid *et al.* (1995) studied the problem of the coherent combination of probability forecasts. Briefly, a *coherent combination formula* is a function $f: [0, 1]^k \rightarrow [0, 1]$ such that

$$f(p_1, \dots, p_k) = \mathbb{P}(Y = 1 | p_1, \dots, p_k) \quad (2)$$

under some joint distribution of the random vector $(Y, p_1, \dots, p_k)$. The conditional probability property (2) implies calibration. Thus, part (a) of theorem 1 shows, for any $k \geq 2$, that any non-trivial linear combination formula with non-negative coefficients is incoherent. This generalizes a result of Dawid *et al.* (1995), page 275, which applies in the case of $k = 2$ sources.

Theorem 2 of Wallsten and Diederich (2001) considers the combination of expert probability judgements, assuming that the assessments are conditionally independent and that each expert's expressed (overt) opinion is a monotone stochastic transform of a hidden (covert) opinion, which is calibrated. Then the arithmetic mean of the expert opinions becomes increasingly diagnostic of the future event as the number of experts grows to ∞ , roughly in the sense that if the mean exceeds $\frac{1}{2}$ the true conditional event probability converges to 1 and otherwise converges to 0. Consequently, the calibration curve for the arithmetic mean of the expert opinions becomes sigmoidal with a fixed point at $\frac{1}{2}$. In contrast with our theorem 1, which is a finite sample result and does not make any assumptions on the dependence structure, Wallsten and Diederich (2001) relied critically on the asymptotic scenario and conditional independence.

Another related result is theorem 4.1 of Genest and Schervish (1985), which adopts a Bayesian point of view and derives a formula for the posterior opinion of a decision maker. Similarly to Wallsten and Diederich's (2001) findings, the true posterior opinion converges to 1, or 0, if the individual judgements lie above, or below, $\frac{1}{2}$. This result also depends on the conditional independence of the individual probability assessments.

Despite theorem 1 being critical of the linear opinion tool, there is overwhelming empirical evidence that linearly combined probability forecasts outperform individual forecasts. This is not a contradiction and can readily be explained, by noting that linear opinion pools outperform individual forecasts, but are suboptimal themselves and can potentially be improved on by using non-linear recalibration methods.

2.2. Simulation study

We now illustrate our theoretical findings in a simulation study. First we describe a statistical

model that gives rise to a joint distribution for the binary event Y and probability forecasts p_1 and p_2 , which represent forecasters with access to independent sources of information. Then we define linearly combined forecasts and assess calibration.

Specifically, let

$$p = \Phi(a_1 + a_2),$$

where $a_1 \sim \mathcal{N}(0, 1)$ and $a_2 \sim \mathcal{N}(0, 2)$ are independent random variables and Φ denotes the standard normal cumulative distribution function. Suppose that Y is a Bernoulli random variable with conditional success probability

$$\mathbb{P}(Y = 1|p) = \mathbb{E}(Y|p) = p.$$

Forecaster 1 has access to a_1 only. This assessor's *probability forecast* p_1 is the conditional event probability

$$p_1 = \mathbb{P}(Y = 1|a_1) = \mathbb{E}(Y|a_1) = \mathbb{E}(p|a_1) = \mathbb{E}\{\Phi(a_1 + a_2)|a_1\} = \Phi(a_1/\sqrt{3}). \quad (3)$$

The second forecaster has knowledge of source a_2 only, whence *probability forecast* p_2 becomes

$$p_2 = \mathbb{P}(Y = 1|a_2) = \Phi(a_2/\sqrt{2}). \quad (4)$$

The final equality stems from the fact that if $X \sim \mathcal{N}(\mu, \sigma^2)$ then $\mathbb{E}\{\Phi(X)\} = \Phi\{\mu/\sqrt{(\sigma^2 + 1)}\}$, which is proved in Appendix A. Evidently, p , p_1 and p_2 are calibrated.

We take p_1 and p_2 as the individual forecasts from which we form combinations, namely the equally weighted linear opinion pool (which we shall denote as the ELP), i.e. the equally weighted average of p_1 and p_2 , and an optimally weighted linear opinion pool (which we shall denote as the OLP). The OLP weights for p_1 and p_2 are estimated on a training sample of size 10000, using the maximum likelihood method and the special case of the log-likelihood function (10) below, in which $\alpha = \beta = 1$. Table 1 shows the OLP estimates and their standard errors. The second individual forecast p_2 , which resolves events and non-events more successfully, obtains a substantially higher OLP weight w_2 of about $\frac{3}{4}$.

In the simulation experiment, we consider an independent test sample of size 10000 from the joint distribution of Y , p_1 and p_2 and generate the combined ELP and OLP forecasts. Fig. 1 shows empirical calibration curves or reliability diagrams (Sanders, 1963; Pocerlich, 2009) for the four types of forecasts, which plot the conditional empirical event frequency against the forecast probability. The circles show the conditional empirical frequency; the broken curves give pointwise 95% lower and upper critical values under the null hypothesis of calibration, obtained with the bootstrap technique of Bröcker and Smith (2007). Significant deviations from the diagonal suggest that a forecast is uncalibrated. The inset histograms show the frequency distribution of the forecast probabilities and can be used diagnostically to assess sharpness.

Table 1. Maximum likelihood estimates of OLP and BLP parameters in the simulation example of Sections 2.2 and 3.2, with standard errors in parentheses

<i>Method</i>	w_1	w_2	α
OLP	0.246 (0.014)	0.754 (0.014)	
BLP	0.519 (0.005)	0.481 (0.005)	9.55 (0.35)

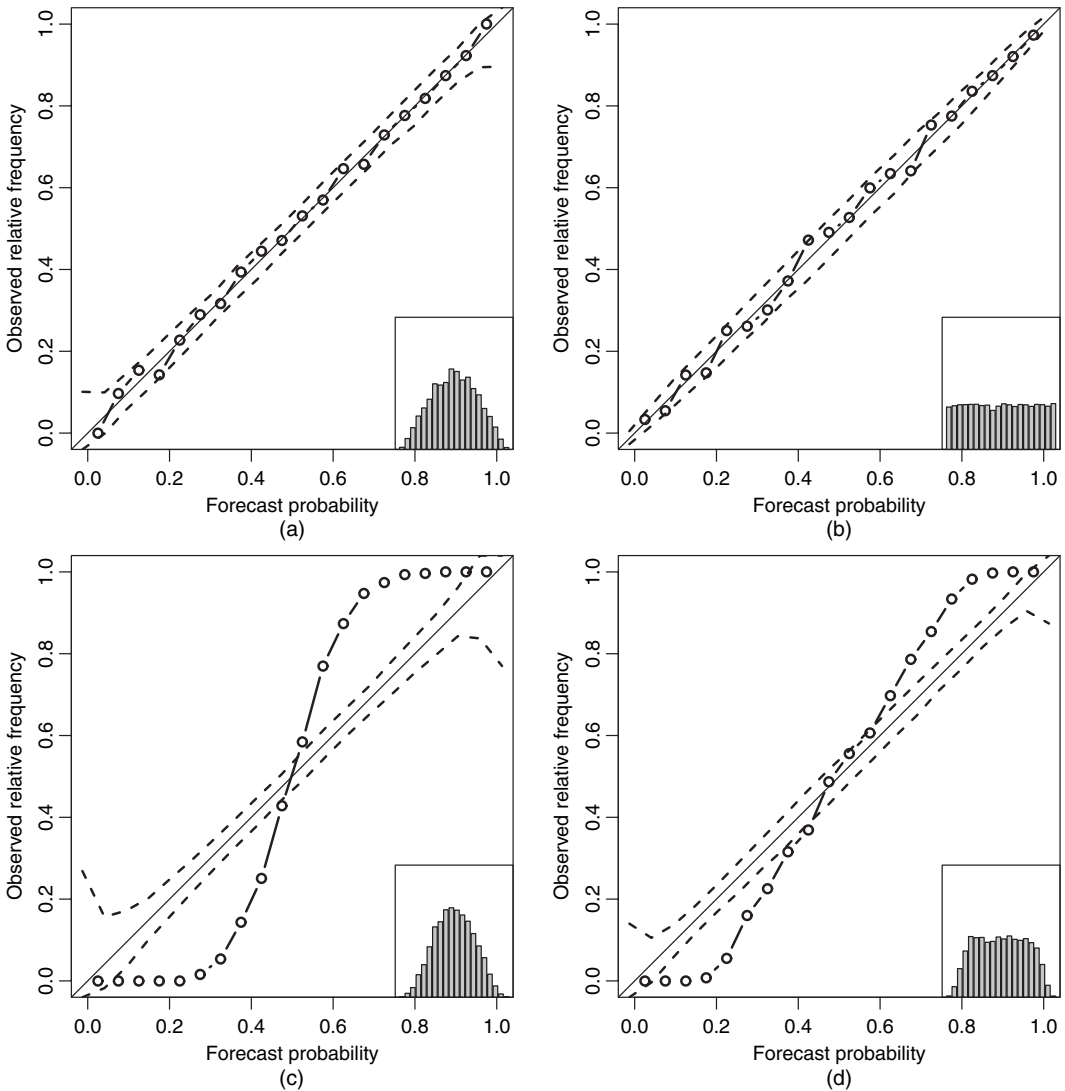


Fig. 1. Calibration curves and 95% bootstrap intervals under the null hypothesis of calibration for the individual and linearly combined forecasts in the simulation example of Section 2.2 (the histograms show the empirical distribution of the forecast values over the unit interval): (a) forecast p_1 ; (b) forecast p_2 ; (c) ELP; (d) OLP

The calibration curves for the individual forecasts p_1 and p_2 show that they are empirically well calibrated, and the inset histograms confirm that p_2 is the sharper forecast, with forecast probabilities that are further away from the climatological event frequency, $p_0 = \frac{1}{2}$. The linearly pooled ELP and OLP forecasts are empirically uncalibrated. The direction of departure is as expected, towards underconfidence, and the extent of the lack of calibration is startling, even for the optimally weighted OLP forecast.

3. Recalibration

We have seen that the linear opinion pool yields a suboptimal combined probability forecast,

in that it is uncalibrated even in the ideal case in which the individual sources are calibrated. If the individual forecasts are uncalibrated, the need for recalibration typically is even more pronounced. Before proposing a method that addresses these issues, by applying a recalibration transform to the linear opinion pool, we digress to discuss a theoretically optimal approach to forecast aggregation.

We have chosen to work in a probabilistic setting that considers the joint distribution of the binary event and the individual probability forecasts. In this framework, the theoretically optimal combined forecast $\hat{p}$ is the conditional probability (CP), or conditional expectation of the binary event Y , given the individual forecasts $p_1, \dots, p_k$, i.e.

$$\hat{p} = \mathbb{P}(Y = 1 | p_1, \dots, p_k) = \mathbb{E}(Y | p_1, \dots, p_k). \quad (5)$$

By definition, this is the best approximation of the binary random variable Y in terms of the individual probability forecasts $p_1, \dots, p_k$, in the sense that $\mathbb{E}(\hat{p} - Y)^2 \leq \mathbb{E}(p - Y)^2$ for all functions p that are measurable with respect to the σ -algebra that is generated by $p_1, \dots, p_k$. Hence, $\hat{p}$ minimizes the expected Brier score and, indeed, the expectation of any strictly proper scoring rule S , in that

$$\begin{aligned} \mathbb{E}\{S(\hat{p}, Y)\} &= \mathbb{E}[\mathbb{E}\{S(\hat{p}, Y) | p_1, \dots, p_k\}] \\ &= \mathbb{E}\{\hat{p} S(\hat{p}, 1) + (1 - \hat{p}) S(\hat{p}, 0)\} \\ &\leq \mathbb{E}\{\hat{p} S(p, 1) + (1 - \hat{p}) S(p, 0)\} \\ &= \mathbb{E}[\mathbb{E}\{S(p, Y) | p_1, \dots, p_k\}] \\ &= \mathbb{E}\{S(p, Y)\}, \end{aligned}$$

with equality if and only if $p = \hat{p}$ almost surely. Under the conditions of theorem 1, the conditional probability $\hat{p}$ is a necessarily non-linear function of the individual forecasts, except for some special cases in which it is linear with at least one coefficient being negative (Dawid *et al.*, 1995). In the simulation study of Section 2.2 there are two individual forecasts, p_1 and p_2 , and the conditional probability (5) equals

$$\begin{aligned} \hat{p} &= \mathbb{P}(Y = 1 | p_1, p_2) \\ &= \mathbb{P}(Y = 1 | a_1, a_2) \\ &= \Phi(a_1 + a_2) \\ &= \Phi\{\Phi^{-1}(p_1)\sqrt{3} + \Phi^{-1}(p_2)\sqrt{2}\}. \end{aligned} \quad (6)$$

This is of the generalized linear form (24) of Dawid *et al.* (1995) with a normal quantile link function.

3.1. The beta-transformed linear opinion pool

In the practice of forecasting, the functional form of the conditional probability (5) is unknown and needs to be estimated from training data. Non-parametric approaches are feasible; however, we prefer parsimonious, yet flexible parametric approximations. Specifically, our preferred approach to aggregating individual probability forecasts, $p_1, \dots, p_k$, is first to form a linear opinion pool, and then to apply a beta transform to achieve calibration. We call this the BLP, which takes the form

$$p = H_{\alpha, \beta} \left(\sum_{i=1}^k w_i p_i \right), \quad (7)$$

where $w_1, \dots, w_k \geq 0$ and $w_1 + \dots + w_k = 1$, and

$$H_{\alpha, \beta}(x) = B(\alpha, \beta)^{-1} \int_0^x t^{\alpha-1} (1-t)^{\beta-1} dt \quad \text{for } x \in [0, 1]$$

is the cumulative distribution function of the beta density with shape parameters $\alpha > 0$ and $\beta > 0$. Note that the BLP model nests the traditional linear opinion pool that arises in the special case when $\alpha = \beta = 1$. If furthermore $w_1 = \dots = w_k = 1/k$ we recover the ELP. Although the use of the beta transform for the purpose of calibration dates back at least to Graham (1996), the statistical model (7) that merges the linear opinion pool with a parametric recalibration transformation appears to be new. It applies very generally and can be used to aggregate calibrated as well as uncalibrated sources.

In many cases, full generality in equation (7) may not be needed or desirable. For instance, it is often useful to assume that the recalibration transform $H_{\alpha, \beta}$ satisfies

$$\begin{aligned} H_{\alpha, \beta}(x) &\leq x & \text{for } x \leq x_0, \\ H_{\alpha, \beta}(x) &\geq x & \text{for } x \geq x_0 \end{aligned} \quad (8)$$

for some $x_0 \in (0, 1)$. This can be enforced by putting conditions on α and β . For example, if the individual forecasts are calibrated, theorem 1 suggests that the linear opinion pool is underconfident, in the sense that its calibration curve lies under the diagonal for small forecast probabilities, and above the diagonal for high probabilities, with a fixed point at some $x_0 \in (0, 1)$. The aforementioned results of Wallsten and Diederich (2001) support the choice of $x_0 = \frac{1}{2}$, under which condition (8) can be enforced by requiring that

$$\alpha = \beta \geq 1. \quad (9)$$

If we aim to address the hard–easy effect that has been described in the psychological literature (Lichtenstein *et al.* (1982) and Kynn (2008), page 253), the fixed point in condition (8) can be taken to be $x_0 = \frac{3}{4}$.

We now describe how we go about parameter estimation for the BLP model (7). Suppose that $y_1, \dots, y_n$ are binary observations in the training set. Let $p_{i1}, \dots, p_{in}$ denote the respective individual probability forecasts, for sources $i = 1, \dots, k$. The aggregated BLP forecast then takes the form

$$p_t = H_{\alpha, \beta} \left(\sum_{i=1}^k w_i p_{it} \right) \quad \text{for } t = 1, \dots, n,$$

where the index ranges over the instances in the training set. Assuming independence, the log-likelihood function for the BLP model (7) can be expressed as

$$\begin{aligned} l(w_1, \dots, w_k; \alpha, \beta) &= \sum_{t=1}^n \{y_t \log(p_t) + (1 - y_t) \log(1 - p_t)\} \\ &= \sum_{t=1}^n y_t \log \left\{ H_{\alpha, \beta} \left(\sum_{i=1}^k w_i p_{it} \right) \right\} + \sum_{t=1}^n (1 - y_t) \log \left\{ 1 - H_{\alpha, \beta} \left(\sum_{i=1}^k w_i p_{it} \right) \right\}. \end{aligned} \quad (10)$$

We obtain maximum likelihood estimates of the weights $w_1, \dots, w_k$ and the recalibration parameters α and β by numerically optimizing the log-likelihood function (10) under the constraints that $w_1, \dots, w_k \geq 0$, $w_1 + \dots + w_k = 1$, $\alpha > 0$ and $\beta > 0$. As noted above, it is often appropriate to enforce further constraints, with inequality (9) being one such example. The traditional,

non-transformed linear opinion pool arises when $\alpha = \beta = 1$. Estimated standard errors can be obtained in the usual way, by inverting a numerical approximation to the Hessian of the log-likelihood function at the maximum likelihood estimates. The estimates can also be interpreted as optimum score estimates based on the logarithmic scoring rule, in the sense described by Gneiting and Raftery (2007), page 375. The latter interpretation does not rely on any assumption of independence, and our results in concert with those of Wilks (1991) suggest robustness to non-independence.

3.2. Simulation study revisited

We return to the simulation study in Section 2.2 and fit the BLP to combine the individual probability forecasts p_1 and p_2 . Then we compare the result with the theoretically optimal forecast, the CP forecast (5) which here has the closed form solution (6).

Recall that both p_1 and p_2 are calibrated, so, as we explained in the previous section, we estimate the BLP model (7) under constraint (9), i.e. we assume that $\alpha = \beta \geq 1$. Table 1 shows maximum likelihood estimates for the BLP parameters and compares them with the respective OLP estimates. The individual forecasts p_1 and p_2 receive approximately equal BLP weights, which is much in contrast with the OLP model. The estimate for the BLP recalibration parameter α is far from the identity transform that arises when $\alpha = 1$, reflecting the striking lack of calibration of the traditional linear opinion pool.

Have we succeeded in our goal of approximating the theoretically optimal CP forecast (6) by the estimated, non-linearly aggregated BLP model (7)? The empirical calibration curve for the BLP forecast in Fig. 2 does not show any systematic departure from the diagonal, and the inset histogram shows that it is much sharper than any of the individual or linearly combined forecasts. A more detailed analysis reveals that if $0 < p_1 = p_2 < 1$ the maximal difference between the CP forecast and the fitted BLP model is 0.0215.

Table 2 shows the mean Brier or quadratic score and its reliability, resolution and uncertainty components for the various forecasts (Murphy, 1973; Dawid, 1986). Suppose that the

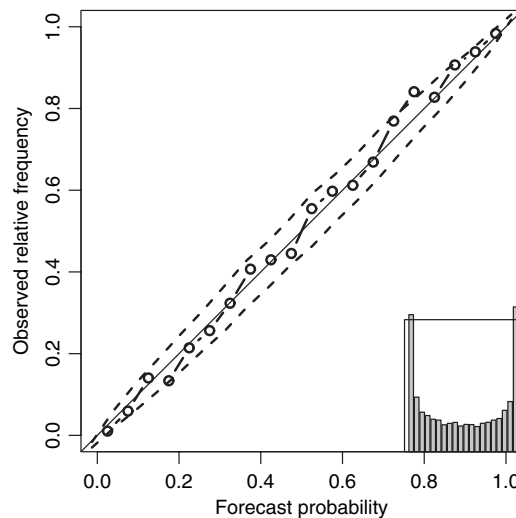


Fig. 2. Calibration curve and 95% bootstrap intervals under the null hypothesis of calibration for the BLP forecast in the simulation example of Sections 2.2 and 3.2, for the same independent test sample as that used before (the histogram shows the empirical distribution of the forecast values over the unit interval)

Table 2. Out-of-sample mean Brier score BS and its reliability REL, resolution RES and uncertainty UNC components for the probability forecasts in the simulation example of Sections 2.2 and 3.2

Forecast	BS	REL	RES	UNC
p_1	0.2094	0.0002	0.0408	0.2500
p_2	0.1657	0.0004	0.0847	0.2500
ELP	0.1563	0.0418	0.1354	0.2500
OLP	0.1531	0.0120	0.1088	0.2500
BLP	0.1137	0.0005	0.1368	0.2500
CP	0.1126	0.0003	0.1377	0.2500

probability forecasts p_t for the binary event y_t , where $t = 1, \dots, n$, take discrete values $f_i \in [0, 1]$, where $i = 1, \dots, I$. Let n_i be the number of times that the forecast value f_i occurs, so that $n = n_1 + \dots + n_I$, and let q_i be the respective empirical conditional event frequency, i.e. the *ex post* recalibrated forecast. Let

$$\bar{q} = \frac{1}{n} \sum_{i=1}^I n_i q_i = \frac{1}{n} \sum_{t=1}^n y_t$$

denote the marginal event frequency. Then the mean Brier score,

$$BS = \frac{1}{n} \sum_{t=1}^n (p_t - y_t)^2,$$

decomposes as

$$BS = \underbrace{\frac{1}{n} \sum_{i=1}^I n_i (f_i - q_i)^2}_{\text{REL}} - \underbrace{\frac{1}{n} \sum_{i=1}^I n_i (q_i - \bar{q})^2}_{\text{RES}} + \underbrace{\bar{q}(1 - \bar{q})}_{\text{UNC}}.$$

The reliability term REL quantifies calibration and is negatively oriented, i.e. the smaller the better. The resolution component RES equals the variance of the *ex post* recalibrated forecast and is positively oriented. For a calibrated forecast, it quantifies sharpness; for an uncalibrated forecast, it measures potential sharpness. As noted above, we generally seek a forecast which is as sharp as possible subject to its being calibrated (Murphy and Winkler, 1987; Gneiting *et al.*, 2007). The uncertainty term UNC is computed from the observations alone and is independent of the forecast. If the probability forecast is a continuous variable, the decomposition depends on a binning of the forecast values and is approximate only. It can be made exact by considering two additional components in the decomposition, as proposed by Stephenson *et al.* (2008). In our case, the extra terms make very little difference, and we consider the classical decomposition only.

From Table 2 we see that the linearly combined ELP and OLP forecasts have lower Brier score than any of the individual forecasts. In both cases, the improvement stems from the resolution component, which is high, because the *ex post* recalibrated forecast is sharp, even though the forecast itself is uncalibrated and lacks sharpness, as reflected in Fig. 1. The BLP forecast is much better calibrated, and simultaneously more successful in resolving events and non-events, than the ELP and OLP forecasts, resulting in a hugely improved Brier score. As expected the

theoretically optimal CP forecast shows the lowest Brier score. However, the BLP forecast is a very close competitor; it is nearly as well calibrated and nearly as sharp as the CP forecast.

3.3. Uncalibrated components

In the above simulation experiment, each individual source was calibrated, and we fitted the BLP model (7) under the constraint that $\alpha = \beta$. However, the BLP approach is more general and applies equally in situations in which one or more of the component forecasts are uncalibrated. Furthermore, it can be beneficial to allow for the full BLP model with general parameters $\alpha > 0$ and $\beta > 0$.

In the remainder of the section, we give an example that covers these two types of situation. Specifically, we consider forecast combinations of $p_1 = \Phi(a_1/\sqrt{3})$, defined as previously, and of the probability forecast p_2^* , which we take to be

$$p_2^* = \Phi(1/5 + a_2/2). \quad (11)$$

As illustrated in Fig. 3, the source p_2^* is uncalibrated and its marginal distribution is skewed. Table 3 shows maximum likelihood estimates for OLP and BLP models which have been fitted on the same training sample (i.e. by using the same random seed for a_1 and a_2) as in Sections 2.2 and 3.2. Note that we refer to the constrained BLP model (with $\alpha = \beta$) as symmetric, and to the full model (with general parameters $\alpha > 0$ and $\beta > 0$) as asymmetric.

Table 3. Maximum likelihood estimates of OLP and BLP parameters in the simulation example of Section 3.3, with standard errors in parentheses

Method	w_1	w_2	α	β
OLP	0.265 (0.017)	0.735 (0.017)		
BLP (symmetric)	0.473 (0.005)	0.527 (0.005)	10.11 (0.36)	10.11 (0.36)
BLP (asymmetric)	0.454 (0.005)	0.546 (0.005)	13.72 (0.49)	11.66 (0.42)

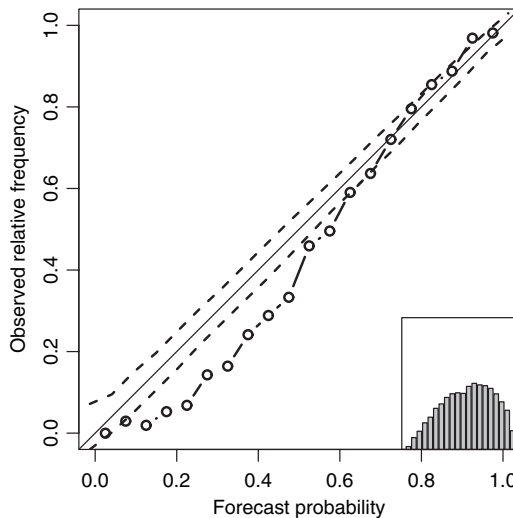


Fig. 3. Calibration curve and 95% bootstrap intervals under the null hypothesis of calibration for the uncalibrated source p_2^* in the simulation example of Section 3.3 (the histogram shows the empirical distribution of the forecast values over the unit interval)

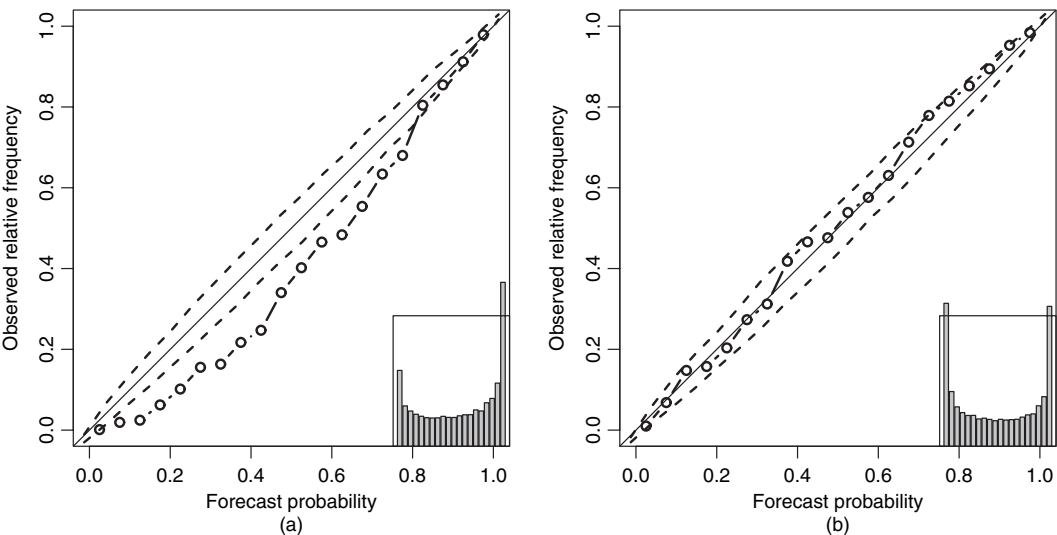


Fig. 4. Calibration curves and 95% bootstrap intervals under the null hypothesis of calibration for the BLP forecasts in the simulation example of Section 3.3 (the histograms show the empirical distribution of the forecast values over the unit interval): (a) symmetric BLP forecast; (b) asymmetric BLP forecast

Table 4. Out-of-sample mean Brier score BS and its reliability REL, resolution RES and uncertainty UNC components for the probability forecasts in the simulation example of Section 3.3

Forecast	BS	REL	RES	UNC
p_1	0.2094	0.0002	0.0408	0.2500
p_2^*	0.1740	0.0084	0.0844	0.2500
ELP	0.1671	0.0514	0.1343	0.2500
OLP	0.1641	0.0321	0.1179	0.2500
BLP (symmetric)	0.1215	0.0084	0.1369	0.2500
BLP (asymmetric)	0.1132	0.0005	0.1373	0.2500
CP	0.1126	0.0003	0.1377	0.2500

Fig. 4 and Table 4 show performance results for the same independent test sample of size 10000 as that used before. The calibration curves in Fig. 4 demonstrate that the symmetric BLP forecast is uncalibrated, whereas the asymmetric BLP forecast is empirically calibrated. The Brier scores in Table 4 confirm that the linearly combined ELP and OLP forecasts outperform each of the individual sources p_1 and p_2^* . However, the non-linearly combined BLP forecasts show much better predictive performance. The asymmetric general version of the BLP forecast outperforms the symmetric version, which is uncalibrated, and performs nearly as well as the theoretically optimal CP forecast.

4. Case-study: probability of precipitation forecasts

We turn to a data example on statistical and NWS probability of precipitation forecasts in continental USA. With about a third of the US economy being weather sensitive, and severe weather causing billions of dollars in damage and hundreds of deaths annually, there is a critical need

for calibrated and sharp probabilistic weather forecasts, to allow for optimal decision-making under inherent uncertainty (Dutton, 2002; Regnier, 2008).

Baars and Mass (2005) considered probability of precipitation forecasts for 29 meteorological stations at major urban centres spread across continental USA. They compared the performance of individual and linearly combined model output statistics (MOS) and NWS forecasts and concluded that a linear opinion pool of the machine-generated MOS forecasts is competitive with or superior to the NWS forecast at nearly all locations. Here we consider the aggregate performance of individual and combined forecasts at all 29 stations, based on the automated ‘GMOS’, ‘EMOS’ and ‘NMOS’ forecasts (Baars and Mass, 2005), and the human-generated, operational NWS forecast. The MOS probability forecasts are statistical forecasts that apply

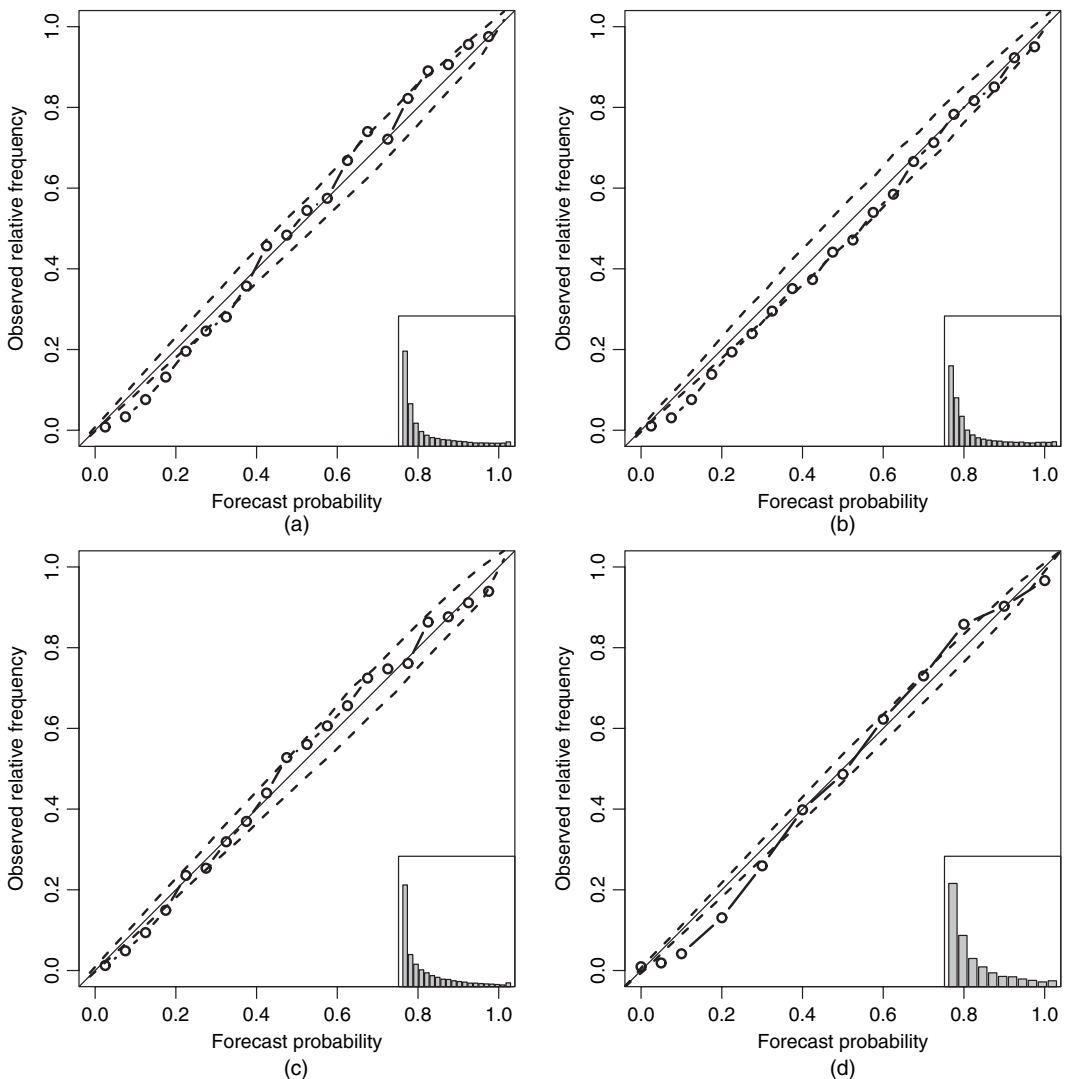


Fig. 5. Calibration curves and 95% bootstrap intervals under the null hypothesis of calibration for the four individual probability of precipitation forecasts in the test period (the histograms show the empirical distribution of the forecast values over the unit interval): (a) GMOS; (b) EMOS; (c) NMOS; (d) NWS

regression techniques to the output of a numerical weather prediction model and recent weather observations (Glahn and Lowry, 1972; Wilks, 2006). The MOS forecasts are recorded in multiples of a hundredth; the NWS forecasts come in multiples of 1/10th, except that a forecast probability of 0.05 is issued occasionally.

We consider the 2-days-ahead probability of precipitation forecasts for the 12-h term called period 2 by Baars and Mass (2005), with our data ranging from July 1st, 2003, to March 3rd, 2008. This includes but is not limited to the 1-year record that was studied by Baars and Mass (2005). We use the first two years (from July 1st, 2003, to June 30th, 2005) as training data, on which we fit OLP and BLP models that apply at all stations simultaneously. The balance of the record (from July 1st, 2005, to March 3rd, 2008) is used as test data on which we evaluate the forecasts. All results are aggregated over the test period and the 29 stations.

Fig. 5 shows calibration curves for the four individual forecasts over the test period. We are in the desirable situation in which the calibration curves show only minor deviations from the diagonal, and so we fit the BLP model (7) under constraint (9). Hence, the BLP model has a single additional recalibration parameter, $\alpha \geq 1$, when compared with the traditional linear opinion pool.

4.1. Combining statistical forecasts

Following Baars and Mass (2005), we consider combined probability forecasts that use the three statistical probability forecasts, namely the GMOS, EMOS and NMOS forecasts. As previously, the ELP is obtained as the simple average of the three forecasts. Table 5 shows maximum likelihood estimates for the OLP and the BLP, which we fit on the training data. For both methods, the GMOS and EMOS weights are about equal and nearly reach $\frac{1}{2}$, with the NMOS weight being much smaller. This is unsurprising, because NMOS is the oldest system and is well known to be the least accurate of the forecasts considered. The maximum likelihood estimate of the BLP recalibration parameter α is 1.48.

Reliability diagrams for the combined forecasts are shown in Fig. 6. The calibration curve for the OLP forecast deviates significantly from the diagonal; the effect is stronger than for any of the individual forecasts, and the direction of the departure agrees with our theoretical results, in that the linearly combined forecast is underconfident. The calibration curve for the ELP forecast is very similar and so it is not shown here. The non-linearly recalibrated BLP forecast is empirically well calibrated and sharper than the OLP forecast.

Table 6 shows the Brier score and its reliability, resolution and uncertainty components for the individual and combined forecasts. The BLP forecast performs the best, both in terms of the Brier score, the reliability or calibration component and the resolution component. The

Table 5. Combined probability forecasts in the precipitation example, using the statistical forecasts only†

Method	Estimates from the following forecasts:			α
	GMOS	EMOS	NMOS	
OLP	0.485 (0.026)	0.465 (0.027)	0.050 (0.020)	
BLP	0.462 (0.022)	0.447 (0.022)	0.091 (0.021)	1.48 (0.03)

†Maximum likelihood estimates for the OLP and BLP parameters from the training period with standard errors in parentheses.

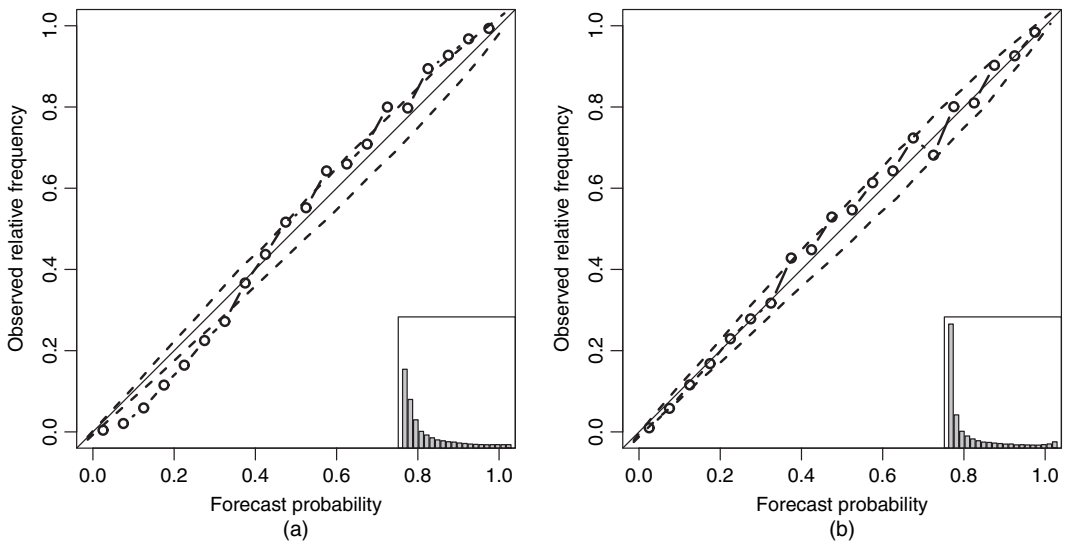


Fig. 6. Calibration curves and 95% bootstrap intervals under the null hypothesis of calibration for the OLP and BLP probability of precipitation forecasts in the test period, using the statistical forecasts only (the histograms show the empirical distribution of the forecast values over the unit interval): (a) OLP; (b) BLP

Table 6. Mean Brier score BS and its reliability REL, resolution RES and uncertainty UNC components for individual and combined probability of precipitation forecasts in the test period, using the statistical forecasts only

Forecast	BS	REL	RES	UNC
GMOS	0.0815	0.0011	0.0739	0.1543
EMOS	0.0866	0.0011	0.0688	0.1543
NMOS	0.0934	0.0005	0.0614	0.1543
ELP	0.0814	0.0023	0.0751	0.1543
OLP	0.0800	0.0021	0.0764	0.1543
BLP	0.0783	0.0004	0.0764	0.1543

improvement of the non-linear BLP method over the linear OLP forecast is about the same as that of the OLP forecast over the best individual source, the GMOS forecast.

4.2. Combining statistical and National Weather Service forecasts

We turn to combined probability forecasts that are based on all four individual sources, now including the NWS forecast, in addition to the GMOS, EMOS and NMOS forecasts. This possibility was not explored by Baars and Mass (2005), who aimed to compare the automated MOS forecasts with the subjective, human-generated NWS forecast.

Table 7 shows maximum likelihood estimates for the OLP and BLP models, which we fit on the training data. For both methods, the GMOS and EMOS forecasts receive weights that are nearly equal, at about 0.37. The NWS forecast receives weights of 0.27 and 0.22 for the OLP and BLP methods respectively; the weights for the NMOS forecast are negligible. The maximum likelihood estimate of the BLP recalibration parameter α is 1.49.

Calibration curves for the OLP and BLP forecasts are shown in Fig. 7. We see the now familiar

Table 7. Combined probability forecasts in the precipitation example, using all four individual forecasts, including the NWS forecast†

Method	Estimates from the following forecasts:				α
	GMOS	EMOS	NMOS	NWS	
OLP	0.362 (0.031)	0.368 (0.030)	0.000 (0.026)	0.270 (0.032)	1.49 (0.03)
BLP	0.371 (0.024)	0.377 (0.023)	0.032 (0.022)	0.220 (0.024)	

†Maximum likelihood estimates for the OLP and BLP parameters from the training period with standard errors in parentheses.

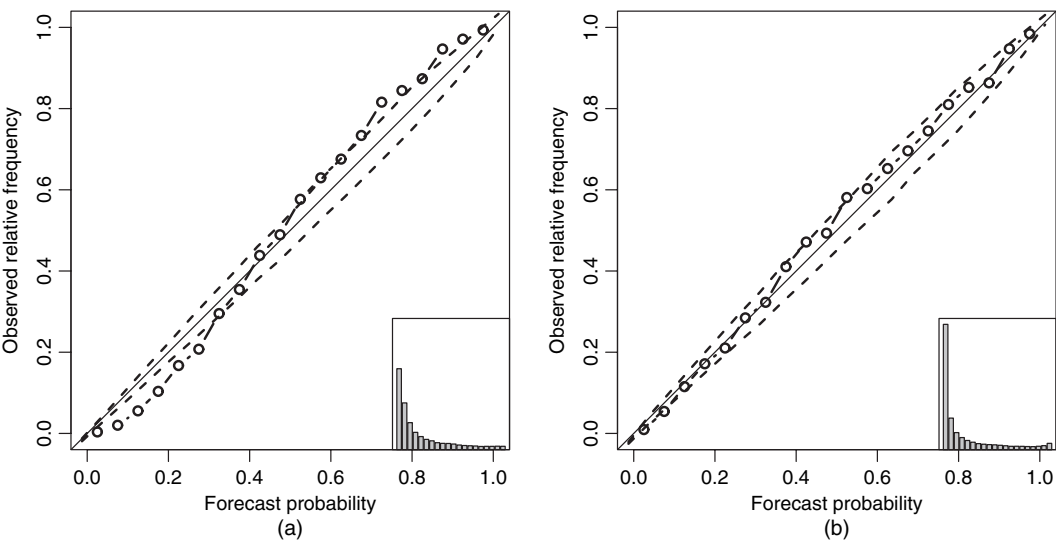


Fig. 7. Calibration curves and 95% bootstrap intervals under the null hypothesis of calibration for the OLP and BLP probability of precipitation forecasts in the test period, using all four individual forecasts, including the NWS forecast: (a) OLP; (b) BLP

Table 8. Mean Brier score BS and its reliability REL, resolution RES and uncertainty UNC components for individual and combined probability of precipitation forecasts in the test period, using all four individual forecasts, including the NWS forecast

Forecast	BS	REL	RES	UNC
GMOS	0.0815	0.0011	0.0739	0.1543
EMOS	0.0866	0.0011	0.0688	0.1543
NMOS	0.0934	0.0005	0.0614	0.1543
NWS	0.0827	0.0009	0.0725	0.1543
ELP	0.0800	0.0026	0.0770	0.1543
OLP	0.0789	0.0024	0.0778	0.1543
BLP	0.0770	0.0004	0.0777	0.1543

pattern, in that the linearly combined OLP forecast lacks calibration. The BLP forecast is empirically well calibrated, and considerably sharper than the OLP forecast. The Brier scores in Table 8 echo these results. The BLP forecast outperforms the OLP and ELP forecasts, which perform better than any of the individual forecasts. If we compare the results with Table 6, we see that the combined probability forecasts benefit from the inclusion of the human-generated NWS forecast, with the improvement due to an increase in resolution.

5. Discussion

Our aim in this paper is to provide theoretical and applied guidance in combining probability forecasts from distinct, calibrated or uncalibrated sources. Historically, the linear opinion pool has been the preferred method for doing this. Indeed, there is overwhelming empirical evidence that linearly combined probability forecasts perform better than individual forecasts, and our results make no exception. That said, our paper demonstrates theoretically and empirically that the linear opinion pool is suboptimal, lacking both calibration and sharpness. To address these shortcomings, we propose the use of the non-linearly recalibrated BLP that nests the traditional, linearly combined probability forecast.

Theorem 1 is our analytic key result; it shows that the linear opinion pool is uncalibrated, even in the desirable case in which the individual sources are calibrated. This is a finite sample result that does not make any restrictive assumptions about the joint dependence structure of the individual forecasts, and it complements the asymptotic results of Wallsten and Diederich (2001) that rely on an assumption of conditional independence. It would be of great interest to bridge the finite sample and asymptotic scenarios, and to establish a more general result, roughly to the extent that linearly combined probability forecasts are uncalibrated and underconfident, resulting in probability statements that are closer to the naive climatological forecast than necessary. A result of this type could perhaps be formulated for a general class of averaging operators and under a minimal assumption of marginal consistency, in lieu of calibration.

Empirically, the shortcomings of the linear opinion pool have been well documented in an interdisciplinary strand of literature that includes the works of Clemen and Winkler (1987), Winkler and Poses (1993), Vislocky and Fritsch (1995), Ariely *et al.* (2000), Wallsten and Diederich (2001) and Johnson *et al.* (2001). Despite their ubiquity, these issues have frequently been overlooked, with some of our own work (Sloughter *et al.*, 2007) being one such example. Indeed, Fig. 7 of Sloughter *et al.* (2007) shows the typical S-shaped calibration curve for a linearly combined, underconfident probability forecast, even though the effect is comparably small.

With a view towards applied forecasting problems, we recommend a transition from the traditional linear opinion pool to the non-linearly recalibrated BLP. The BLP model (7) has at most two, and typically only one, additional parameters when compared with the linear opinion pool, and it is easy to fit, using the maximum likelihood method or related optimum score techniques. More general and more complex, parametric or non-parametric approaches to the aggregation of probability forecasts can easily be envisioned, including but not limited to copula models (Jouini and Clemen, 1996), and might provide effective approximations to the hypothetical, ideally combined forecast, namely the CP forecast (5). However, more complex statistical models bear the danger of overfitting, and the resulting gains in predictive performance, if any, are likely to be incremental.

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Appendix A: Mathematical details

A.1. Proof of theorem 1

From the basic properties of Bernoulli random variables and conditional expectations,

$$\mathbb{P}(Y = 1) = \mathbb{E}(Y^2) = \mathbb{E}(Y) = \mathbb{E}\{\mathbb{E}(Y|p)\} = \mathbb{E}(p) = \mathbb{E}(q),$$

which will be used repeatedly in what follows. We first prove part (a). For a contradiction, suppose that p is calibrated, i.e. $p = q$ almost surely. Then we can condition on p to see that

$$\mathbb{E}(Y - p)^2 = \mathbb{E}\{p(1 - p)\}. \quad (12)$$

We proceed to show that under the conditions of the theorem equality in expression (12) is violated. For this, note that

$$\begin{aligned} \mathbb{E}(Y - p)^2 &= \mathbb{E}\left\{\sum_{i=1}^k w_i(Y - p_i)\right\}^2 \\ &= \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}\{(Y - p_i)(Y - p_j)\} \\ &= \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}(Y - p_i Y - p_j Y + p_i p_j) \\ &= \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}\{\mathbb{E}(Y|p_i) - \mathbb{E}(p_i Y|p_i) - \mathbb{E}(p_j Y|p_j) + p_i p_j\} \\ &= \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}(p_i - p_i^2 - p_j^2 + p_i p_j) \\ &= \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}\{p_i(1 - p_j)\} - \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}(p_i - p_j)^2 \end{aligned}$$

and

$$\mathbb{E}\{p(1 - p)\} = \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}\{p_i(1 - p_j)\},$$

so that

$$\mathbb{E}(Y - p)^2 = \mathbb{E}\{p(1 - p)\} - \sum_{i=1}^k \sum_{j=1}^k w_i w_j \mathbb{E}(p_i - p_j)^2. \quad (13)$$

The double sum on the right-hand side of equation (13) is strictly positive, whence equation (12) is violated, for the desired contradiction.

We now prove part (b). From equation (13) we see that $\mathbb{E}(Y - p)^2 < \mathbb{E}\{p(1 - p)\}$. A straightforward conditioning argument shows that

$$\mathbb{E}(Y - p)^2 = \mathbb{E}(Y - q)^2 + \mathbb{E}(q - p)^2 > \mathbb{E}\{q(1 - q)\}.$$

Hence,

$$\mathbb{E}\{q(1 - q)\} < \mathbb{E}(Y - p)^2 < \mathbb{E}\{p(1 - p)\},$$

which implies that $\mathbb{E}(p^2) < \mathbb{E}(q^2)$. From this, part (b) follows.

As for part (c),

$$\begin{aligned}\mathbb{E}\{S(q, Y)\} &= \mathbb{E}[\mathbb{E}\{S(q, Y)|p\}] \\ &= \mathbb{E}\{q S(q, 1) + (1 - q) S(q, 0)\} \\ &< \mathbb{E}\{q S(p, 1) + (1 - q) S(p, 0)\} \\ &= \mathbb{E}[\mathbb{E}\{S(p, Y)|p\}] \\ &= \mathbb{E}\{S(p, Y)\}\end{aligned}$$

with the inequality being strict, because S is a negatively oriented strictly proper scoring rule and $q = \mathbb{E}(Y|p) \neq p$ with positive probability.

A.2. Details for equations (3) and (4)

The final equality in equation (3) stems from the fact that if $X \sim \mathcal{N}(\mu, \sigma^2)$ then

$$\begin{aligned}\mathbb{E}\{\Phi(X)\} &= \int_{-\infty}^{\infty} \Phi(x) \frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) dx \\ &= \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^x \phi(y) dy \right\} \frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) dx \\ &= \int_{y \leq x} \phi(y) \frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right) dx dy \\ &= \mathbb{P}(Y \leq X) = \mathbb{P}(Y - X \leq 0) = \Phi\left\{ \frac{\mu}{\sqrt{(\sigma^2 + 1)}} \right\},\end{aligned}$$

where ϕ denotes the standard normal density function and Y is standard normal and independent of X , so that $Y - X \sim \mathcal{N}(-\mu, \sigma^2 + 1)$. The conditional distribution of $a_1 + a_2$ given a_1 is normal with mean a_1 and variance 2, whence

$$\mathbb{E}\{\Phi(a_1 + a_2)|a_1\} = \Phi(a_1/\sqrt{3}).$$

An almost identical calculation applies to equation (4).

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